

AMUTHEEZAN SIVAGNANAM

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Visa Status: U.S. Lawful Permanent Resident

[Website](#) | [LinkedIn](#) | [Google Scholar](#)

Nationality: Sri Lanka

Location: State College, PA

SUMMARY

Researcher and **Engineer** with a **PhD** in Informatics, bringing over **six** years of academic research and **two** years of industrial software development experience. Skilled in developing **scalable reinforcement learning** and **deep learning** methods for **optimization** and **decision-making** in complex systems. Strong publication record (**h-index: 7**) in top-tier venues. Proficient in **Python**, **TensorFlow**, **PyTorch**, and **AWS**, with experience in **Agile** and **CI/CD** practices for robust ML model **development** and **integration**.

EDUCATION

Pennsylvania State University

Ph.D., Informatics

Aug 2022 - Aug 2025

Dissertation: *Application of Deep Reinforcement Learning to Solve Optimization Problems in Transportation Domains*

University of Houston (Transferred to Pennsylvania State University)

Ph.D., Computer Science

Aug 2019 - Aug 2022

University of Houston

M.S., Computer Science

Aug 2019 - Aug 2022

University of Moratuwa

B.S., (Hons) Engineering (Computer Science and Engineering)

Jan 2014 - Jan 2018

Final Year Project: *Sentimental Analysis of Twitter using Semi-Supervised Approaches*

RESEARCH INTERESTS

Reinforcement Learning, Optimization, Operational Research, Cyber-Physical Systems

SKILLS

- **Languages:** Python, C/C++, Java
- **AI/ML Libraries:** PyTorch, TensorFlow, Keras, Hugging Face, Scikit-learn
- **Optimization Tools:** CPLEX, Gurobi, Mosek, Google OR-Tools
- **Data processing:** pandas, Spark, matplotlib
- **Miscellaneous Libraries:** Scipy, NumPy, Ray, RLLib, OpenAI Gym
- **Databases:** SQLite, MySQL, OracleDB, MongoDB
- **Deployment:** Git, Docker, AWS (SageMaker, EC2, Lambda, S3)

RESEARCH EXPERIENCE

Pennsylvania State University
Applied Artificial Intelligence Lab

Graduate Research Assistant
Aug 2022 - May 2025

- Lead research projects funded by the **U.S. Department of Energy (DOE)** and the **National Science Foundation (NSF)**, introducing **cost-and energy-efficient** solutions to tackle real-world problems
- Analyzed current state-of-the-art and identified **research gaps** in prior works, leading to the development of **innovative algorithms** for **real-time decision-making**
- Introduce novel **problem formulations** and **mathematical models** to address real-world challenges, while ensuring inherent spatio-temporal and resource constraints
- Proposed **artificial intelligence-based solutions** to tackle **challenging** real-world problems and successfully **deployed** them in relevant **industries**
- Published **research findings** in **AI/ML conferences (ICML)** and assisted in preparing the slides for presenting results at **DOE and NSF meetings**

The University of Houston
Resilient Networks and Systems Lab

Graduate Research Assistant
Sep 2019 - Aug 2022

- Spearheaded research projects funded by the **U.S. Department of Energy (DOE)** and the **National Science Foundation (NSF)**, leading to advancements in **energy-efficient** technologies
 - Focus on understanding real-world **decision-making problems** and identified **gaps** in existing solution approaches, leading to the development of **novel methodologies** for **enhanced decision-making**
 - Propose novel **problem formulations** and **mathematical models** and formulated **problem statements** to effectively address the research limitations in prior research efforts
 - Applied **artificial intelligence-based solution** approaches to **real-world** problems and successfully **deployed** them in relevant **industries**
 - Published **research findings** in **AI conferences (AAAI, IJCAI)** and assisted preparing **slides** for presenting results at **DOE and NSF meetings**
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INDUSTRY EXPERIENCE

LSEG Technology

Post Trade Team

Software Engineer

Jan 2018 - Jul 2019

- Developed application software following **object-oriented** design and development principles, enhancing **code maintainability** and **scalability**
- Followed **agile development practices**, such as **scrum**, to improve **team collaboration** and **project delivery speed**
- Developed software solutions using **Java**, **Python**, and **C++**, improving **system performance** and **reliability**
- Modified SQL-queries, tested against **BDD**, ensuring correctness of **system functionality**
- Enhanced **back-end regression** to achieve proper code coverage in based on unit-testing, ensuring **compatibility** and reducing **testing time**
- Developed a new **report generation system** for assessing the state of current testing framework, **automating email** at the end of regression
- Contributed to **code integration** and **deployment plans**, ensuring **seamless software updates** and minimizing **downtime**
- Participated in **professional training** programs conducted by Millennium IT Software and Post Trade Team
- Worked on front-end development for both **product** and **solution** which consists of **enhancement**, **bug fixing**, **merging** and **introducing new features**

WSO2 Lanka PVT Ltd

Data Analytics Team

Software Engineering Intern

Jul 2016 – Dec 2016

- Developed an HL7 Monitoring Solution by integrating **WSO2 ESB**, **DAS**, and **BAM**, enabling real-time and batch processing of healthcare data
 - Developed an alert generation system analyzing descriptive **HL7/FHIR** data to monitor **disease outbreaks**, triggering timely email and SMS notifications
 - Created **Spark scripts** for batch analytics and **Siddhi execution plans** for real-time alerts, including disease outbreak and patient wait-time notifications
 - Engineered a mechanism to evaluate **hospital functionality** by assessing **admission and discharge** messages, including bed and oxygen cylinder availability
 - Implemented interactive dashboards using **Jaggery**, **JavaScript**, **jQuery**, **Leaflet.js**, and **DataTables**, featuring gadgets like charts, maps, and tables
 - Utilized **HAPI test panel** to simulate HL7 v2 messages and validate the monitoring pipeline end-to-end
 - Packaged the entire monitoring solution as a **Carbon Application (CApp)**, bundling event streams, receivers, publishers, and visualization artifacts
 - Attended workshops and gained hands-on experience with **Git**, **MSF4J (Microservices for Java)**, and **WSO2 product architecture** for enterprise middleware solutions
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PUBLICATIONS

CONFERENCE PROCEEDINGS

- [C1] **Sivagnanam, A.**, Pettet, A., Lee, H., Mukhopadhyay, A., Dubey, A., & Laszka, A. (2024). Multi-agent reinforcement learning with hierarchical coordination for emergency responder stationing. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*.
- [C2] **Sivagnanam, A.**, Kadir, S. U., Mukhopadhyay, A., Pugliese, P., Dubey, A., Samaranayake, S., & Laszka, A. (2022, July). Offline vehicle routing problem with online bookings: A novel problem formulation with applications to paratransit. In *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence (IJCAI-22)* (pp. 3933–3939).
- [C3] **Sivagnanam, A.**, Ayman, A., Wilbur, M., Pugliese, P., Dubey, A., & Laszka, A. (2021, May). Minimizing energy use of mixed-fleet public transit for fixed-route service. In *Proceedings of the AAAI Conference on Artificial Intelligence (Vol. 35, No. 17, pp. 14930–14938)*.
- [C4] Sen, R., **Sivagnanam, A.**, Laszka, A., Mukhopadhyay, A., & Dubey, A. (2024). Grid-aware charging and operational optimization for mixed-fleet public transit. In *Proceedings of the 2024 IEEE 27th International Conference on Intelligent Transportation Systems (ITSC)* (pp. 4172–4179). IEEE.
- [C5] Atefi, S., **Sivagnanam, A.**, Ayman, A., Grossklags, J., & Laszka, A. (2023). The benefits of vulnerability discovery and bug bounty programs: Case studies of Chromium and Firefox. In *Proceedings of the ACM Web Conference 2023 (WWW '23)* (pp. 2209–2219). Association for Computing Machinery.
- [C6] Pavia, S., Rogers, D., **Sivagnanam, A.**, Wilbur, M., Edirimanna, D., Kim, Y., Mukhopadhyay, A., Pugliese, P., Samaranayake, S., Laszka, A., & Dubey, A. (2024). SmartTransit.AI: A dynamic paratransit and microtransit application. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI '24)* (Article No. 1028, pp. 8767–8770).
- [C7] Ayman, A., Wilbur, M., **Sivagnanam, A.**, Pugliese, P., Dubey, A., & Laszka, A. (2020). Data-driven prediction of route-level energy use for mixed-vehicle transit fleets. In *Proceedings of the 2020 IEEE International Conference on Smart Computing (SMARTCOMP)* (pp. 41–48). IEEE.

JOURNALS

[J1] Wilbur, M., Ayman, A., **Sivagnanam, A.**, Ouyang, A., Poon, V., Kabir, R., Vadali, A., Pugliese, P., Freudberg, D., Laszka, A., & Dubey, A. (2023). Impact of COVID-19 on public transit accessibility and ridership. *Transportation Research Record*, 2677(4), 531–546.

[J2] Ayman, A., **Sivagnanam, A.**, Wilbur, M., Pugliese, P., Dubey, A., & Laszka, A. (2021). Data-driven prediction and optimization of energy use for transit fleets of electric and ICE vehicles. *ACM Transactions on Internet Technology*, 22(1), Article 7, 1–29.

WORKSHOPS

[W1] **Sivagnanam, A.**, Atefi, S., Ayman, A., Grossklags, J., & Laszka, A. (2021). On the benefits of bug bounty programs: A study of Chromium vulnerabilities. In *Workshop on the Economics of Information Security (WEIS)* (Vol. 10).

BOOK CHAPTERS

[BC1] Wilbur, M., **Sivagnanam, A.**, Ayman, A., Samaranayake, S., Dubey, A., & Laszka, A. (2023, August). Artificial intelligence for smart transportation. In **Y. Vorobeychik & A. Mukhopadhyay (Eds.)**, *Artificial Intelligence and Society* (book chapter). ACM Press.

PROJECTS

PHD RESEARCH PROJECTS

Dynamic Vehicle Routing with Advance Booking

Oct 2023 – Aug 2025

- Focus on a real-world micro-transit operations where riders may **book** rides in few hours **advance** with but expect **acceptance or rejection** at booking time
- Developed a **Deep Reinforcement Learning**(DRL)–based policy that learns to distinguish the high quality route plans that could increase the service rates
- Integrated an **anytime** VRP solver to periodically **enhance** the quality of routes **between** bookings, enabling enough room for future requests in the limited vehicles
- Achieved **100% service rate** and outperform other competitive state-of-the-art dynamic VRP baselines using **MCVRP** and **Google OR Tools** and **Rolling Horizon**, as demonstrated through experiments on **real-world** data from mid-size U.S. cities

Emergency Responder Stationing [[Code](#)]

Nov 2021 – June 2024

- Developed a novel multi-agent DRL with **hierarchical coordination** to address the **resource reallocation problem** in the context of **emergency responder management**
- Utilized the Deep Deterministic Policy Gradient (**DDPG**) algorithm to train agents in different hierarchy (i.e., **city-scale redistribution** (high-level) and **region-scale reallocation** (low-level))

- Incorporated a **Transformer-based** actor network to handle **variable numbers of responders** in region-scale reallocation
- Leveraged **min-cost flow** (city-level) and **max-weight matching** (region-level) to ensure precise and **feasible mapping** from continuous to discrete actions, while preserving **gradient flow** during training
- Integrated **low-level critics** to provide **reward feedback** to high-level agents, enhancing **training stability** and **performance**
- Achieved **1000× faster** decision-making and **reduced response delays by 5–13 seconds** on **real-world** datasets from Nashville, TN and Seattle, WA

The Benefits of Vulnerability Discovery and Bug Bounty Programs *Feb 2020 – May 2023*

- Collect and compose the publicly available **chromium data** using **Monorail API**, **Google Release Notes** and **Google Chrome Hall of Fame**
- Perform intensive **data cleaning** process to **identify** the original reporters (achieve an **accuracy of 98%**), **duplicates** issues, and time at which the issue got **patched** and **released** to public
- Demonstrated that bug-bounty programs **complement** the efforts internal security teams by uncovering a diverse range of vulnerability types from external bug-hunters
- Delivered **actionable insights** to enhance bug-bounty effectiveness, including **targeted guidance** for bug hunters toward vulnerabilities with risk of real-world exploitation potential

Offline Vehicle Routing Problem with Online Bookings [[Code](#)] *Feb 2020 – Jul 2022*

- Focus on a real-world paratransit operations where riders **book** trips in **advance** with **time flexibility** but expect **tight pickup windows** at booking time
- Introduced a **novel problem formulation** that blends the **scalability** of offline VRP with the **real-time responsiveness** of dynamic VRP
- Developed a **DRL-based policy** that learns to assign optimal time windows under demand uncertainty and booking-time constraints
- Integrated an **anytime** VRP solver to incrementally **enhance** the quality of route plans **between** bookings, enabling better utilization of limited vehicles
- Achieved at-least **20% cost reduction** over baseline methods using **Google OR Tools** and **VRoom** with naive window assignments, as demonstrated through extensive experiments on **real-world** data from paratransit-operations of Chattanooga, TN

Minimizing Energy Use of Mixed-Fleet Public Transit [[Code](#)] *Aug 2019 – May 2021*

- Formulated a **mathematical model** to **optimize** energy consumption for public transit agencies operating mixed fleets of electric vehicles (EVs) and internal combustion engine vehicles (ICEVs)
- Transformed the model into an Mixed Integer Linear Programming (MILP) formulation to obtain **exact** solutions for **small** to **medium-sized** problem instances
- Developed a **scalable** solution approach combining **heuristics** and **metaheuristics** (e.g., **local search**, **genetic algorithms**) to **efficiently** solve large-scale instances in polynomial time
- Introduced heuristics and metaheuristics **perform reasonably** well close to **MILP**, and the **difference cap at 60%**

- Reduced **annual energy costs** by **\$140K** for the Chattanooga public transit agency through the proposed approach

UNDERGRAD RESEARCH PROJECT

Sentimental Analysis of Twitter using Semi-Supervised Approaches *Nov 2016 – Nov 2017*

- Developed a **sentiment analysis** model to classify tweets identify the polarity of tweets as **positive, negative, and neutral**
- Utilized **semi-supervised learning techniques**, including self-training, co-training, to address **limited and noisy labeled data**
- Initiated training with a small **manually-labeled** dataset, and iteratively expanded it by **automatically labeling** based on high-confidence data (i.e., tweets)
- Among training strategies **co-training** results in **superior** results in terms of F1-score

UNDERGRAD INTERNSHIP PROJECT

HL7 Monitoring System *July 2016 – Dec 2016*

- Built an end-to-end **HL7/FHIR monitoring** system using WSO2 ESB, DAS, and BAM for both **real-time (Siddhi)** and **batch (Spark)** analytics of healthcare data
- Designed an **alert** system to detect disease **outbreaks** and **long patient wait times**, delivering **email** and **SMS** notifications based on HL7/FHIR data streams
- Engineered hospital **functionality assessments** by analyzing admission/discharge events, tracking resources like **bed** and **oxygen cylinder availability**
- Developed **interactive** dashboards with Jaggery, JavaScript, Leaflet.js, and DataTables, and packaged the solution as a **WSO2 Carbon Application (CApp)** for easy deployment

TALKS

IJCAI (Vienna, Austria) 2022

For the paper: Offline Vehicle Routing Problem with Online Bookings:
A Novel Problem Formulation with Applications to Paratransit

WEIS (Virtual Workshop) 2021

For the workshop paper: On the Benefits of Bug Bounty Programs:
A Study of Chromium Vulnerabilities

AAAI (Virtual Conference) 2021

For the paper: Minimizing Energy Use of Mixed-Fleet Public Transit
for Fixed-Route Service

SERVICES

Program Committee Member

International Joint Conference on Artificial Intelligence (AI For Social Good)	2025
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Reviewer

International Joint Conference on Artificial Intelligence (AI For Social Good)	2024
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Reviewer

AI4Research Workshop @ IJCAI2024	2024
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Auxiliary Reviewer

22nd International Conference on Autonomous Agents and Multiagent Systems	2023
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AWARDS AND HONORS

Scholarship

2022/08 - 2025/08

Graduate Tuition Fellowship, Pennsylvania State University
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Scholarship

2019/08 - 2022/08

Graduate Tuition Fellowship, University of Houston
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Information Security Quiz 2015 - Runners Up

2015

Issued by Sri Lanka Computer Emergency Readiness Team (SLCERT) and Information and Communication Technology Agency of Sri Lanka (ICTA)

Mahapola Scholarship

2014 - 2017

REFERENCES

Dr. Aron Laszka (PhD Supervisor)

Position: **Assistant Professor**

Affiliation: **Pennsylvania State University**

Email: aql5923@psu.edu

Phone: **814-865-1551**

Website: <https://aronlaszka.com>

Dr. Abhishek Dubey

Position: **Associate Professor**

Affiliation: **Vanderbilt University**

Email: abhishek.dubey@vanderbilt.edu

Phone: **615-322-8775**

Website: <https://abhishekdubey.bio>

Dr. Weidong Shi

Position: **Associate Professor**

Affiliation: **University of Houston**

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